

et al.). Ignoring $\sigma_{\mathcal{S}_1}$ means that, for instance, this approach would choose to simply scale $k \rightarrow \infty$, even though this does not change the test at all. Even when this is not possible, Sutherland et al. (2017) found this approach notably worse than maximizing (4); we confirm this in our experiments.

MMD-GANs (Li et al., 2017; Binkowski et al., 2018) also simply maximize $\widehat{\text{MMD}}_u$ to identify the differences between their model $\mathbb{Q}_\theta$ and target $\mathbb{P}$. If $\mathbb{Q}_\theta$ is quite far from $\mathbb{P}$, however, an MMD-GAN requires a “weak” kernel to identify a path for improving $\mathbb{Q}_\theta$ (Arbel et al., 2018), while our ideal kernel is one which perfectly distinguishes $\mathbb{P}$ and $\mathbb{Q}_\theta$ and would likely give no signal for improvement. Our algorithm, theoretical guarantees, and empirical evaluations thus all differ significantly from those for MMD-GANs.

3. Limits of Simple Kernels

We can use the criterion $\hat{J}_\lambda$ of (4) even to select parameters among a simple family, such as the lengthscale of a Gaussian kernel. Doing so on the *Blob* problem of Figure 1 illustrates the limitations of using MMD with these kernels.

In Figure 2c, we show how the maximal value of $\hat{J}$ changes as we see more samples from $\mathbb{P}$ and $\mathbb{Q}$, for both a family of Gaussian kernels (green dashed line) and a family (1) of deep kernels (red line). The optimal $\hat{J}$ is always higher for the deep kernels; as expected, the empirical test power (Figure 2a) is also higher for deep kernels.

Most simple kernels used for MMD tests, whether the Gaussian we use here or Laplace, inverse multiquadric, even automatic relevance determination kernels, are all translation invariant: $k(x, y) = k(x - t, y - t)$ for any $t \in \mathbb{R}^d$. (All kernels used by Sutherland et al. (2017), for instance, were of this type.) Hence the kernel behaves the same way across space, as in Figure 1b. This means that for distributions whose behavior varies through space, whether because principal directions change (as in Figure 1) so the shape should be different, or because some regions are much denser than others and so need a smaller lengthscale (e.g. Wenliang et al., 2019, Figures 1 and 2), any single global choice is suboptimal.

Kernels which are not translation invariant, such as the deep kernels (1) shown in Figure 1c, can adapt to the different shapes necessary in different areas.

4. Relationship to Classifier-Based Tests

Another popular method for conducting two-sample tests is to train a classifier between $S_\mathbb{P}^{tr}$ and $S_\mathbb{Q}^{tr}$, then assess its performance on $S_\mathbb{P}^{te}$, $S_\mathbb{Q}^{te}$. If $\mathbb{P} = \mathbb{Q}$, the classification problem is impossible and performance will be at chance.

The most common performance metric is the accuracy

(Lopez-Paz & Oquab, 2017); this scheme is fairly common among practitioners, and Kim et al. (2020) showed it to be optimal in rate, but suboptimal in constant, in one limited setting (linear discriminant analysis between high-dimensional elliptical distributions, e.g. Gaussians, with identical covariances). We will call this approach a Classifier Two-Sample Test based on Sign, C2ST-S. Letting $f : \mathcal{X} \rightarrow \mathbb{R}$ output classification scores, the C2ST-S statistic is $\widehat{\text{acc}}(S_\mathbb{P}, S_\mathbb{Q}; f)$ given by

$$\frac{1}{2n} \sum_{X_i \in S_\mathbb{P}} \mathbb{1}(f(X_i) > 0) + \frac{1}{2n} \sum_{Y_i \in S_\mathbb{Q}} \mathbb{1}(f(Y_i) \leq 0).$$

Let $\text{acc}(\mathbb{P}, \mathbb{Q}; f) := \frac{1}{2} \Pr(f(X) > 0) + \frac{1}{2} \Pr(f(Y) \leq 0)$; $\widehat{\text{acc}}$ is unbiased for acc and has a simple asymptotically normal null distribution.

Although it is perhaps not immediately obvious this is the case, C2ST-S is almost a special case of the MMD. Let

$$k_f^{(S)}(x, y) = \frac{1}{4} \mathbb{1}(f(x) > 0) \mathbb{1}(f(y) > 0). \quad (6)$$

A C2ST-S with f is equivalent to an MMD test with $k_f^{(S)}$:

Proposition 3. *It holds that*

$$\begin{aligned} \text{MMD}(\mathbb{P}, \mathbb{Q}; k_f^{(S)}) &= |\text{acc}(\mathbb{P}, \mathbb{Q}; f) - \frac{1}{2}| \\ \widehat{\text{MMD}}_b(S_\mathbb{P}, S_\mathbb{Q}; k_f^{(S)}) &= |\widehat{\text{acc}}(S_\mathbb{P}, S_\mathbb{Q}; f) - \frac{1}{2}|. \end{aligned}$$

Proof. The mean embedding $\mu_\mathbb{P}$ under $k_f^{(S)}$ is simply $\frac{1}{2} \mathbb{E} \mathbb{1}(f(X) > 0) = \frac{1}{2} \Pr(f(X) > 0)$, so the MMD is

$$\frac{1}{2} \left| \Pr(f(X) > 0) - \Pr(f(Y) > 0) \right| = \left| \text{acc}(\mathbb{P}, \mathbb{Q}; f) - \frac{1}{2} \right|.$$

Moreover, $\widehat{\text{acc}}$ is acc on empirical distributions. $\square$

The C2ST-S, however, selects f to maximize cross-entropy (approximately maximizing $\widehat{\text{acc}}$), while we maximize $\hat{J}_\lambda$ (4). Although $k_f^{(S)}$ is not differentiable, maximizing (3) would exactly maximize acc and hence maximize test power (Lopez-Paz & Oquab, 2017, Theorem 1).

Accessing f only through its sign allows for a simple null distribution, but it ignores f ’s measure of confidence: a highly confident output extremely far from the decision boundary is treated the same as a very uncertain one lying in an area of high overlap between $\mathbb{P}$ and $\mathbb{Q}$, dramatically increasing the variance of the statistic. A scheme we call C2ST-L instead tests difference in means of f on $\mathbb{P}$ and $\mathbb{Q}$ (Chen & Cloninger, 2019). Let

$$k_f^{(L)}(x, y) = f(x)f(y). \quad (7)$$

A C2ST-L is equivalent to an MMD test with $k_f^{(L)}$:

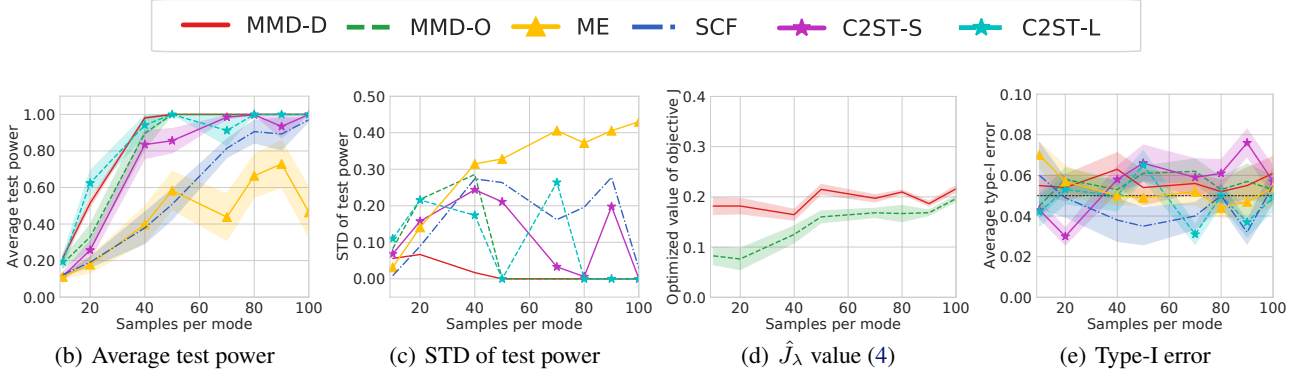


Figure 2. Results on *Blob-S* and *Blob-D* given $\alpha = 0.05$; see Section 7 for details. n_b is the number of samples at each mode, so $n_b = 100$ means drawing 900 samples from each of $\mathbb{P}$ and $\mathbb{Q}$. We report, when increasing n_b , (a) average test power, (b) standard deviation of test power, (c) the value of $\hat{J}_\lambda$, and (d) average type-I error. (a), (b) and (c) are on *Blob-D*, and (d) is on *Blob-S*. Shaded regions show standard errors for the mean, and the black line shows α .

Proposition 4. *It holds that*

$$\begin{aligned} \text{MMD}(\mathbb{P}, \mathbb{Q}; k_f^{(L)}) &= |\mathbb{E} f(X) - \mathbb{E} f(Y)| \\ \widehat{\text{MMD}}_b(S_{\mathbb{P}}, S_{\mathbb{Q}}; k_f^{(L)}) &= \left| \frac{1}{n} \sum_{X_i \in S_{\mathbb{P}}} f(X_i) - \frac{1}{n} \sum_{Y_i \in S_{\mathbb{Q}}} f(Y_i) \right|. \end{aligned}$$

Proof. This kernel’s feature map is $k_f^{(L)}(x, \cdot) = f(x)$. $\square$

Now maximizing accuracy (or a cross-entropy proxy) no longer directly maximizes power. This kernel is differentiable, so we can directly compare the merits of maximizing (4) to maximizing cross-entropy; we will see in Section 7.2 that our more direct approach is empirically superior.

Compared to using $k_f^{(L)}$, however, Section 7.2 shows that learned MMD tests also obtain better performance using kernels like (1). This is analogous to a similar phenomenon observed in other problems by Binkowski et al. (2018) and Wenliang et al. (2019): C2STs learn a full discriminator function on the training set, and then apply only that function to the test set. Learning a deep kernel like (1) corresponds to learning only a powerful *representation* on the training set, and then *still learning* f itself from the test set – in a closed form that makes permutation testing simple.

5. Learning Deep Kernels

Choice of kernel architecture. Most previous work on deep kernels has used a kernel κ directly on the output of a featurization network ϕ_ω , $k_\omega(x, y) = \kappa(\phi_\omega(x), \phi_\omega(y))$. This is certainly also an option for us. Any such k_ω , however, is characteristic if and only if ϕ_ω is injective. If we select our kernel well, this is not really a concern.³ Even so, it

³A characteristic kernel on top of even $\phi_\omega(x) = \omega^\top x$ with a *random* ω will be almost surely consistent (Heller & Heller, 2016), and in general the existence of even one good ϕ_ω for a particular

would be reassuring to know that, even if the optimization goes awry, the resulting test will still be at least consistent. More importantly, it can be helpful in optimization to add a “safeguard” preventing the learned kernel from considering extremely far-away inputs as too similar. We can achieve these goals with the form (1), repeated here:

$$k_\omega(x, y) = [(1 - \epsilon)\kappa(\phi_\omega(x), \phi_\omega(y)) + \epsilon] q(x, y).$$

Here ϕ_ω is a deep network (with parameters ω) that extracts features, and κ is a kernel on those features; we use a Gaussian with lengthscale σ_ϕ , $\kappa(a, b) = \exp\left(-\frac{1}{2\sigma_\phi^2}\|a - b\|^2\right)$. We choose $0 < \epsilon < 1$ and q a Gaussian with lengthscale σ_q .

Proposition 5. *Let k_ω be of the form (1) with $\epsilon > 0$ and q characteristic. Then k_ω is characteristic.*

Learning the deep kernel. The kernel optimization and testing procedure is summarized in Algorithm 1. For larger datasets, or when $n \neq m$, we use minibatches in the training procedure; for smaller datasets, we use full batches. We use the Adam optimizer (Kingma & Ba, 2015). Note that the parameters ϵ , σ_ϕ , and σ_q are included in ω , all parameterized in log-space (i.e. we optimize ϵ' where $\epsilon = \exp(\epsilon')$).

Time complexity. Let E denote the cost of computing an embedding $\phi_\omega(x)$, and K the cost of computing (1) given $\phi_\omega(x)$, $\phi_\omega(y)$. Then each iteration of training in Algorithm 1 costs $\mathcal{O}(mE + m^2K)$, where m is the minibatch size; for the moderate m that fit in a GPU-sized minibatch anyway, the mE term typically dominates, matching the complexity of a C2ST. Testing takes time $\mathcal{O}(nE + n^2K + n^2n_{\text{perm}})$, compared to $\mathcal{O}(nE + nn_{\text{perm}})$ for permutation-based C2STs. In either case, the quadratic factors could if necessary be reduced

$\mathbb{P}$, $\mathbb{Q}$ pair is enough that a perfect optimizer would be able to distinguish the distributions (Arbel et al., 2018, Proposition 1).

Algorithm 1 Testing with a learned deep kernel

Input: S_P, S_Q , various hyperparameters used below;
 $\omega \leftarrow \omega_0; \lambda \leftarrow 10^{-8}$;
 Split the data as $S_P = S_P^{tr} \cup S_P^{te}$ and $S_Q = S_Q^{tr} \cup S_Q^{te}$;
Phase 1: train the kernel parameters ω on S_P^{tr} and S_Q^{tr}
for $T = 1, 2, \dots, T_{max}$ **do**
 $X \leftarrow$ minibatch from $S_P^{tr}; Y \leftarrow$ minibatch from S_Q^{tr} ;
 $k_\omega \leftarrow$ kernel function with parameters ω ; # as in (1)
 $M(\omega) \leftarrow \widehat{\text{MMD}}_u^2(X, Y; k_\omega)$; # using (2)
 $V_\lambda(\omega) \leftarrow \hat{\sigma}_{\mathfrak{H}_1, \lambda}^2(X, Y; k_\omega)$; # using (5)
 $\hat{J}_\lambda(\omega) \leftarrow M(\omega) / \sqrt{V_\lambda(\omega)}$; # as in (4)
 $\omega \leftarrow \omega + \eta \nabla_{\text{Adam}} \hat{J}_\lambda(\omega)$; # maximize $\hat{J}_\lambda(\omega)$
end for
Phase 2: permutation test with k_ω on S_P^{te} and S_Q^{te}
 $est \leftarrow \widehat{\text{MMD}}_u^2(S_P^{te}, S_Q^{te}; k_\omega)$
 for $i = 1, 2, \dots, n_{perm}$ **do**
 Shuffle $S_P^{te} \cup S_Q^{te}$ into X and Y
 $perm_i \leftarrow \widehat{\text{MMD}}_u^2(X, Y; k_\omega)$
 end for
Output: $k_\omega, est, p\text{-value } \frac{1}{n_{perm}} \sum_{i=1}^{n_{perm}} \mathbb{1}(perm_i \geq est)$

with the block estimator approach of Zaremba et al. (2013), at the cost of some test power. In our experiments in Section 7, the overall runtime of our methods was scarcely different from the overall runtime of C2STs.

6. Theoretical Analysis

We now show that optimizing the regularized test power criterion based on a finite number of samples works: as n increases, our estimates converge uniformly over a ball in parameter space, and therefore if there is a unique best kernel, we converge to it. Sutherland et al. (2017) gave no such guarantees; this result allows us to trust that, at least for reasonably large n and if our optimization process succeeds, we will find a kernel that generalizes nearly optimally rather than just overfitting to S^{tr} .

We first state a generic result, then show some choices of kernels, particularly deep kernels (1), satisfy the conditions.

Theorem 6. *Let ω parameterize uniformly bounded kernel functions k_ω in a Banach space of dimension D , with $|k_\omega(x, y) - k_{\omega'}(x, y)| \leq L_k \|\omega - \omega'\|$. Let $\bar{\Omega}_s$ be a set of ω for which $\sigma_{\mathfrak{H}_1}^2(\mathbb{P}, \mathbb{Q}; k_\omega) \geq s^2 > 0$ and $\|\omega\| \leq R_\Omega$. Take $\lambda = n^{-1/3}$. Then, with probability at least $1 - \delta$,*

$$\sup_{\omega \in \bar{\Omega}_s} |\hat{J}_\lambda(S_P, S_Q; k_\omega) - J(\mathbb{P}, \mathbb{Q}; k_\omega)| = \mathcal{O} \left(\frac{1}{s^2 n^{1/3}} \left[\frac{1}{s} + \sqrt{D \log(R_\Omega n) + \log \frac{1}{\delta}} + L_k \right] \right).$$

If there is a unique best kernel ω^ , the maximizer of $\hat{J}_\lambda$ converges in probability to ω^* as $n \rightarrow \infty$.*

A version with explicit constants and more details is given in Appendix A (as Theorem 11 and Corollary 12); the proof is based on uniform convergence of the MMD and variance estimators using an ϵ -net argument.

The following results are shown in Appendix A.4. We first show a result on simple Gaussian bandwidth selection.

Proposition 7. *Suppose each $x \in \mathcal{X}$ has $\|x\| \leq R_X$, and we choose the bandwidth of a Gaussian kernel among a set whose minimum is at least $1/R_\Omega$. Then the conditions of Theorem 6 are met with $D = 1$ and $L_k = 2R_X/\sqrt{e}$.*

Our results also apply to multiple kernel learning, where in fact the exact maximizer of $\hat{J}_\lambda$ is efficiently available (Proposition 27).

Proposition 8. *Let $\{k_i\}_{i=1}^D$ be a fixed set of kernels, with $\sup_x k_i(x, x) \leq K$ for all i . Then picking $k_\omega = \sum_{i=1}^D \omega_i k_i$ among some set of ω with $\sum_{i=1}^D \omega_i^2 \leq R_\Omega^2$ satisfies the conditions of Theorem 6 with $L_k = K\sqrt{D}$.*

We finally establish our results for fully-connected deep kernels; it also applies to convolutional networks with a slightly different R_Ω (Remark 25). The constants in L_k are given in Proposition 23.

Proposition 9. *Take k_ω as in Section 5, with ϕ_ω a fully-connected network with depth Λ and D total parameters, whose activations are 1-Lipschitz with $\sigma(0) = 0$ (e.g. ReLU). Suppose the operator norm of each weight matrix and L_2 norm of each bias vector are at most R_Ω , and each $x \in \mathcal{X}$ has $\|x\| \leq R_X$. Then k_ω meets the conditions of Theorem 6 with dimension D and $L_K = \mathcal{O} \left(\Lambda R_\Omega^{\Lambda-1} \frac{R_X+1}{\sigma_\phi} \right)$.*

The dependence on s in Theorem 6 is somewhat unfortunate, but the ratio structure of J means that otherwise, errors in very small variances can hurt us arbitrarily. Even so, “near-perfect” kernels (with reasonably large MMD and very small variance) will likely still be chosen as the maximizer of the regularized criterion, even if we do not estimate the (extremely large) ratio accurately. Likewise, near-constant kernels (with very small variance but still small J) will generally have their J underestimated, and so are unlikely to be selected when a better kernel is available. The ϵq component in (1) may also help avoid extremely small variances.

Given N data points, this result also gives insight into how many we should use to train the kernel and how many to test. With perfect optimization, Corollary 14 shows a bound on the asymptotic power of the test is maximized by training on $\Theta \left((N\sqrt{\log N})^{\frac{3}{4}} \right)$ points, and testing on the remainder.

7. Experimental Results

7.1. Comparison on Benchmark Datasets

We compare the following tests on several datasets: